

Ankit Kumar

Machine Learning Engineer | Agentic AI, RAG & Production ML Systems

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Seattle, WA | F-1 STEM OPT through May 2028

Machine Learning Engineer with 4+ years building and deploying production ML and GenAI systems across agentic RAG, hybrid retrieval and reranking, physics-informed modeling, time-series forecasting, and embedded inference. Improved top-5 retrieval hit rate from 42% to 86% on a fixed labeled 50-query evaluation set. Owns systems end to end, from ambiguous requirements through evaluation, deployment, and cross-functional validation.

EXPERIENCE

Milwaukee Electric Tool Corporation

July 2025 – July 2026

Machine Learning Engineer – Front End Innovation (FEI)

- Led development of a collocation-based PyTorch PINN thermal-warning feature using a three-body ODE for a chainsaw platform planned for 200K+ units; held maximum per-profile RMSE to 2.2°C across 30 held-out profiles and provided warnings up to 60 seconds before hard shutoff.
- Cut experiment iteration from two hours to 40 minutes by building Azure ETL and training pipelines that synchronized lab and field data, automated preprocessing and model execution, and versioned datasets, parameters, metrics, and artifacts in MLflow.
- Reduced validation time from 30 days to three by validating laboratory load-brake cycles against field-use heat-generation profiles, establishing a repeatable substitute for hands-on cutting tests.
- Identified a sensor-induced 4–5°C model error through ablation and counterfactual testing from –20°C to 40°C, supporting its removal and \$100K in projected hardware savings.
- Partnered with the principal firmware engineer to port the PyTorch model to embedded C and built a time-causal Python reference twin with Streamlit telemetry replay; caught two release-blocking defects in OCV/DCIR handling and initialization before field validation, avoiding an estimated 30+ days of rework.

Mitra | Mason Innovation Exchange, GMU

June 2024 – May 2025

Founder & Lead ML Engineer – Selected University Venture

- Led end-to-end development of an AI-powered university decision-support platform that earned 78% positive ratings in professor-led classroom evaluations involving 200+ students.
- Built an event-driven Scrapy, AWS, and PySpark pipeline processing 8K+ webpages per crawl; filtered approximately 70% of low-value pages before embedding and maintained a source-traceable corpus of 50K+ chunks through 30-minute Glue ETL runs.
- Improved top-5 retrieval hit rate from 42% (21/50) to 86% (43/50) on a fixed labeled 50-query GMU evaluation set by combining MiniLM embeddings, FAISS HNSW, lexical search, weighted reciprocal rank fusion, and cross-encoder reranking.
- Deployed a bounded LangGraph workflow as a Dockerized FastAPI service on ECS Fargate, calling SageMaker-hosted retrieval and reranking endpoints; capped each request at five agent actions and two LLM calls, routed low-evidence cases to retrieval retry, clarification, or abstention, traced agent execution with LangSmith, and monitored services with CloudWatch.
- Raised precision@20 from 25% to 45% (an average of 5 to 9 relevant chunks per query) across two offline cycles by filtering noisy feedback with DistilBERT and retraining a pairwise cross-encoder on hard negatives; gated deployments with Pytest regression tests in GitHub Actions and versioned ECR/S3 artifacts for rollback.

Accenture

February 2021 – July 2023

Packaged App Development Analyst

- Built PySpark and SQL pipelines on Amazon S3 to standardize point-in-time fundamentals for approximately 9,500 public companies into partitioned Parquet, reducing company and competitor-analysis preparation from five business days to 1.5 days—a 70% reduction.
- Developed pooled LSTM and XGBoost models with SARIMA baselines for company-KPI series with sufficient history for training and chronological validation; improved two-year directional accuracy from 64% to 78% in stable regimes and from 52% to 61% during transitions against persistence baselines.
- Fine-tuned FinBERT on approximately 75K timestamped earnings calls, news articles, and analyst reports using forward market-relative price movement labels; achieved 0.76 macro-F1 on a chronological holdout and increased transition-period directional accuracy from 61% to 66% over the structured-only ensemble.

PROJECTS

PyLinguist: Cross-Lingual Python Translation | [GitHub](#)

November 2024 – Present

Research project | Submitted to ACL BEA

- Built a two-stage Python translation system across seven languages using token-aware translation, a curated dictionary of 234 Python terms, and Google Translate followed by GPT-4o or Claude 3.5 Haiku refinement while preserving code syntax and structure.
- Evaluated 70 held-out programs using BLEU, syntax validity, structural preservation, semantic similarity, and bilingual review; GPT-4o achieved a 92.5% composite evaluation score versus 61.5% for the translation-only baseline, with 100% syntax validity.

TECHNICAL SKILLS

Languages & Data: Python, SQL, C, Pandas, NumPy, PySpark, ETL, Parquet

GenAI & Agents: RAG, LangGraph, LangChain, Pydantic, Prompt Engineering, Evidence-Gated Generation, LLM/Agent Evaluation

Retrieval & NLP: Hugging Face Transformers, SentenceTransformers, FAISS HNSW, Hybrid Search, Weighted RRF, Cross-Encoder Reranking, FinBERT, DistilBERT

ML & Deep Learning: PyTorch, scikit-learn, XGBoost, LSTM, SARIMA, Model Fine-Tuning, Time-Series Forecasting, Model Validation, BLEU

Physics & Numerical Modeling: PyTorch ODEs, Collocation-Based Physics-Informed Neural Networks (PINNs), Forward Euler, Trapezoidal (Tustin) Integration

Statistics & Experimentation: Statistical Analysis, Experimental Design, Monte Carlo Simulation, Sensitivity Analysis, Ablation & Counterfactual Testing

Software & Deployment: FastAPI, REST APIs, Docker, Git, Pytest, GitHub Actions, CI/CD, Amazon ECR, Amazon ECS Fargate

Cloud & MLOps: AWS (SageMaker, Lambda, S3, SQS, Glue, Athena, CloudWatch), Azure, MLflow, LangSmith, Streamlit

EDUCATION

George Mason University

August 2023 – May 2025

M.S. in Computer Science – Machine Learning Concentration | GPA: 3.90/4.00

Courses: Mining Massive Datasets with MapReduce, Advanced NLP, Machine Learning, Data Mining, Artificial Intelligence

Achievements: Distinguished Academic Achievement Award, College of Engineering and Computing

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate (MLA-C01)
- AWS Certified Developer – Associate (DVA-C02)